Problem 5

Problem 5

Problem 1 *Suppose a function f satisfies the following conditions:*

(a) $f(0) = 0$ *and* $f'(-4) = f'(2) = f'(10) = 0$

(b) $\lim_{x\to 6} f(x) = -\infty$, *and* $\lim_{x\to +\infty} f(x) = 6$

(c) $f'(x) < 0$ *on* $(-\infty, -4)$, $(2, 6)$, *and* $(10, +\infty)$

(d) $f'(x) > 0$ *on* $(-4, 2)$, *and* $(6, 10)$

(e) $f''(x) > 0$ *on* $(-\infty, 0)$, *and* $(14, +\infty)$

(f) $f''(x) < 0$ *on* $(0, 6)$, *and* $(6, 14)$

(a) *List the* ***interval(s)*** *where the function f is* ***both increasing*** and ***concave UP***.

Explanation. $(-4, 0)$

(b) *List the* ***interval(s)*** *where the function f is* ***both increasing*** and ***concave DOWN***.

Explanation. $(0, 2)$, $(6, 10)$

(c) *List the* ***interval(s)*** *where the function f is* ***both decreasing*** and ***concave UP***.

Explanation. $(-\infty, -4)$, $(14, +\infty)$

(d) *List the **interval(s)** where the function f is **both decreasing** and **concave DOWN**.*

Explanation. $(2, 6), \quad (10, 14)$

(e) *List the **x-coordinates** at which f has a **local minimum**. Write "none" if appropriate.*

Explanation. $x = -4$

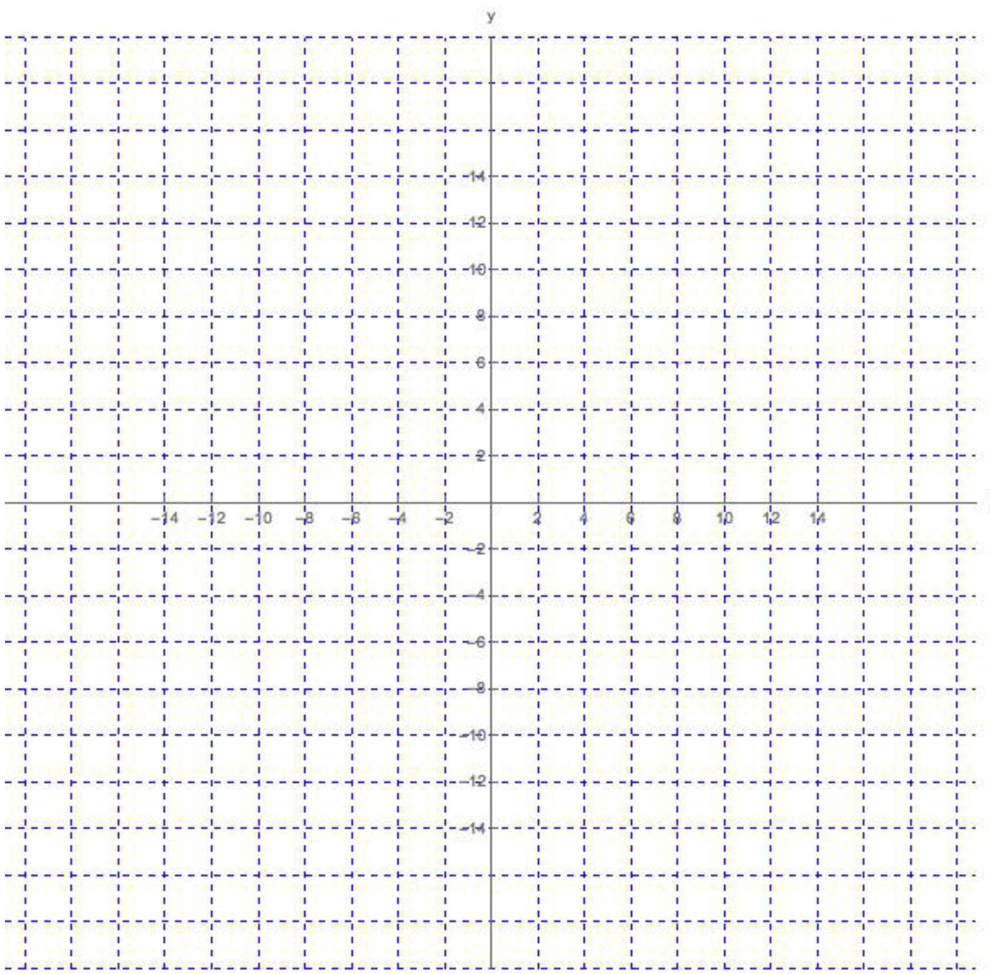
(f) *List the **x-coordinates** at which f has a **local maximum**. Write "none" if appropriate.*

Explanation. $x = 2, \quad x = 10$
(f can also have a local maximum at $x = 6$, if we include $x = 6$ in the domain)

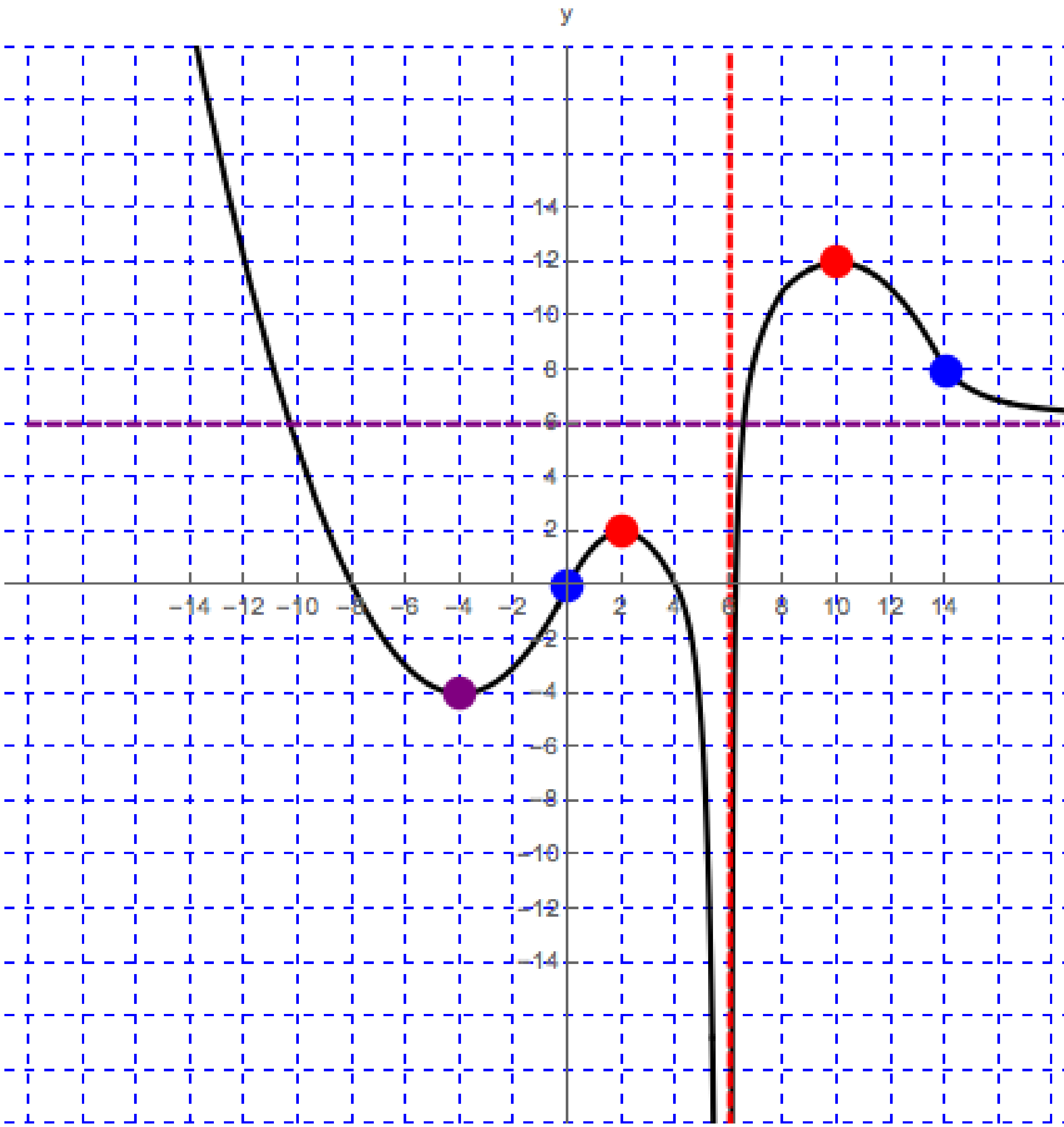
(g) *List the **x-coordinates** of all **inflection points** of f. Write "none" if appropriate.*

Explanation. $x = 0, \quad x = 14$

(h) *Sketch the graph of f.*



Explanation. This is just one of many possible graphs.
The domain of f could be $(-\infty, +\infty)$, and f could satisfy the given conditions.



Purple point : local minimum

Red points : local maxima

Blue points : inflection points